# Chapter 15 — Testing and Debugging

**What you will learn.** Why automated tests pay for themselves and how to write them: `unittest`, then **pytest** — plain `assert`, fixtures, `parametrize`, `tmp_path`, marks, `conftest.py`. Isolating code from slow or external dependencies with **test doubles** and `unittest.mock` (`Mock`, `patch`, `side_effect`, autospec). **Coverage**, the red-green-refactor **TDD** loop, **doctest**, and **property-based testing** with Hypothesis. Then debugging: tracebacks, `pdb` and `breakpoint()`, logging instead of `print`, and a checklist of the most common Python bugs.

**Prerequisites.** Functions, classes, exceptions (Chapters 7, 9, 10). Logging basics: Chapter 14.

**Code.** `chapters/15-testing-and-debugging/code/`: a small module `inventory.py` plus its tests. Install once, run from that folder:

```
pip install pytest pytest-cov hypothesis
cd chapters/15-testing-and-debugging/code
pytest -q                      # 31 passed, 1 skipped, 1 xfailed
python debugging_demo.py
```

("xfailed" = a test *expected* to fail did fail — §15.4.5.)




## 15.1 Why write tests?

### What it is

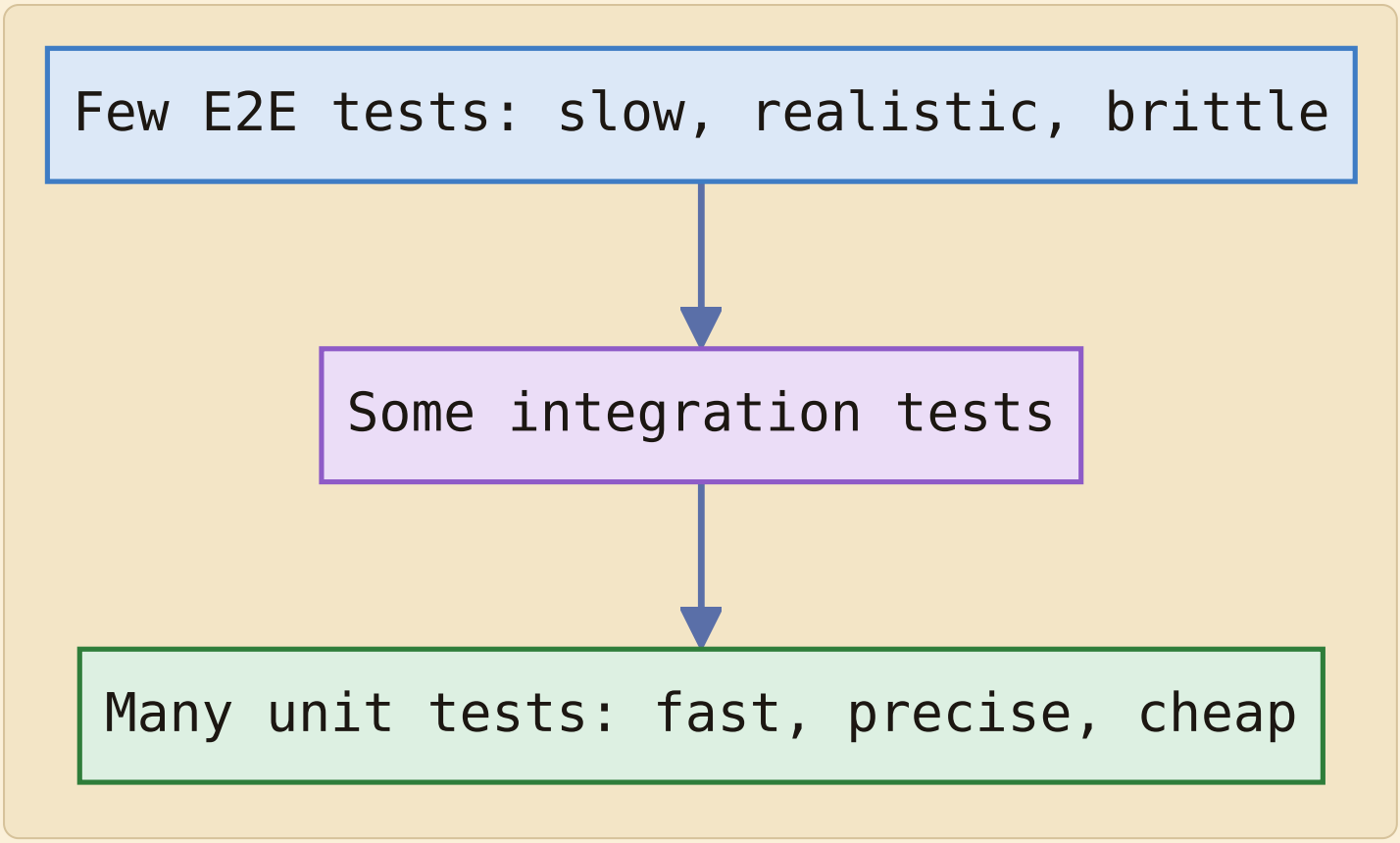
An **automated test** runs your code with known inputs and checks the result.

| Term | Meaning |
|---|---|
| **Unit test** | tests one small piece (a function, a class) in isolation, in milliseconds |
| **Integration test** | tests several pieces together (your code + a real database or file system) |
| **End-to-end (E2E) test** | drives the whole system like a user (HTTP calls, browser) |
| **Test suite** | all the tests of a project |
| **Regression** | a bug that comes back, or a change that breaks something that used to work |
| **SUT** | *system under test* — the code being tested |
| **Arrange–Act–Assert** | the usual three-part shape of a test: set up, do the thing, check the outcome |

### Why it matters

1. **Confidence to change code** — refactoring without tests is guessing; with a good suite you know in seconds if anything broke.
2. **Bugs are cheaper earlier** — minutes on your laptop vs incidents, data fixes and lost trust in production.
3. **Executable documentation** — tests show exactly how code is meant to be used.
4. **Better design** — hard-to-test code is usually too coupled; tests push you to small functions with explicit inputs.
5. Python has **no compiler** to catch a typo in a rarely-run branch — only running the code finds it (type checkers help, Chapter 13).

### How it works — the test pyramid




Mostly fast unit tests, fewer integration tests, a handful of E2E tests for critical paths — all run automatically in **CI** (continuous integration, e.g. GitHub Actions runs `pytest` on every push and pull request).

**What makes a good test**

- **Fast** (milliseconds), **isolated** (independent of other tests and order), **repeatable** (no real clock, network or unseeded randomness), **self-checking** (pass/fail, no human reading output), **focused** (one behaviour, named).
- Test **behaviour, not implementation** — what a function returns or changes, not which private helpers it calls — so refactoring doesn't break tests.

### Real-world examples

CPython's suite (`python -m test`) uses `unittest`; NumPy, pandas, pip and most modern projects use pytest; Django ships a `unittest`-based runner. Their pull requests merge only when CI is green.

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## 15.2 The module under test

Every example tests `inventory.py` (abridged — `# ...` marks omitted lines):

```python
class OutOfStock(Exception):
    """Raised when removing more units than are available."""

def slugify(name: str) -> str:
    """Lower-case, keep letters/digits, join words with '-'.

    >>> slugify("Hello, World!")
    'hello-world'
    """
    words = re.findall(r"[a-z0-9]+", name.lower())
    return "-".join(words)

def apply_discount(price: float, percent: float) -> float:
    if not 0 <= percent <= 100:
        raise ValueError(f"percent must be 0-100, got {percent}")
    return round(price * (1 - percent / 100), 2)

@dataclass
class Inventory:
    stock: dict[str, int] = field(default_factory=dict)

    def add(self, sku: str, qty: int) -> None:
        if qty <= 0:
            raise ValueError("qty must be positive")
        self.stock[sku] = self.stock.get(sku, 0) + qty

    def remove(self, sku: str, qty: int) -> None:
        have = self.stock.get(sku, 0)
        if qty > have:
            raise OutOfStock(f"{sku}: want {qty}, have {have}")
        self.stock[sku] = have - qty

    def low_stock(self, threshold: int) -> list[str]: ...   # built with TDD in §15.8
    def total_units(self) -> int: ...
    def save(self, path: Path) -> None:
        path.write_text(json.dumps(self.stock, sort_keys=True))
    @classmethod
    def load(cls, path: Path) -> "Inventory": ...

def price_in(currency: str, usd: float, client: RateClient) -> float:
    """Convert a USD price using an external rate service (mocked in tests)."""
    rate = client.get_rate(currency)
    if rate <= 0:
        raise ValueError("bad rate")
    return round(usd * rate, 2)
```

In `price_in` the rate service is **passed in** (dependency injection) rather than created inside — which makes it trivial to test with a fake (§15.5).

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## 15.3 `unittest` basics

### What it is

The standard library's test framework, modelled on Java's JUnit (the "xUnit" family). Tests are methods of a `unittest.TestCase` subclass.

### How it works

1. `class TestX(unittest.TestCase)`; each method named `test*` is one test.
2. `setUp()` runs before **each** test, `tearDown()` after each; `setUpClass`/`tearDownClass` (class methods) once per class.
3. **Assertions** (checks that must hold or the test fails): `assertEqual`, `assertTrue`, `assertIn`, `assertIsNone`, `assertAlmostEqual` (7 decimal places by default), `assertRaises` (as a context manager), `assertLogs`.

4. `@unittest.skip(reason)` / `skipIf(cond, reason)`; `@unittest.expectedFailure` for known bugs.
5. Run: `python -m unittest` (auto-discovers `test*.py`), `python -m unittest -v test_inventory_unittest`, or `unittest.main()` in the file.

```
class TestInventory(unittest.TestCase):
    def setUp(self):                      # runs before EACH test method
        self.inv = Inventory()
        self.inv.add("mug", 5)

    def test_add_accumulates(self):
        self.inv.add("mug", 3)
        self.assertEqual(self.inv.stock["mug"], 8)

    def test_remove_too_many_raises(self):
        with self.assertRaises(OutOfStock):
            self.inv.remove("mug", 99)

    def test_discount(self):
        self.assertAlmostEqual(apply_discount(19.99, 10), 17.99)

    @unittest.skip("demo of skipping")
    def test_skipped(self):
        self.fail("never runs")
```

Output of `python test_inventory_unittest.py`:

```
test_add_accumulates (__main__.TestInventory.test_add_accumulates) ... ok
test_discount (__main__.TestInventory.test_discount) ... ok
test_remove_too_many_raises (__main__.TestInventory.test_remove_too_many_raises) ... ok
test_skipped (__main__.TestInventory.test_skipped) ... skipped 'demo of skipping'

----------------------------------------------------------------------
Ran 4 tests in 0.000s

OK (skipped=1)
```

## Trade-offs — unittest vs pytest

| | unittest | pytest |
|---|---|---|
| Install | built in | `pip install pytest` |
| Test shape | class + `self.assertX` methods | plain functions + `assert` |
| Setup | `setUp` / `tearDown` | fixtures (reusable, composable, scoped) |
| Data-driven tests | `self.subTest(...)` loop | `@pytest.mark.parametrize` |
| Failure messages | from the assert method | rewritten `assert` shows values and diffs |
| Plugins (add-on packages) | few | large ecosystem (coverage, xdist for parallel runs, asyncio, django…) |
| Runs the other's tests | — | yes, pytest runs `unittest` tests unchanged |

### When to use it

When you can't add dependencies (stdlib-only tools, CPython itself) or in an existing unittest codebase. Otherwise pytest is the default — and it runs unittest tests, so you can migrate gradually.

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## 15.4 pytest

### What it is

**pytest**, the most widely used Python test framework: plain functions with plain `assert`; it discovers, runs and explains failures in detail.

### How it works — discovery and running

1. Collects files `test_*.py` / `*_test.py`, functions `test_*`, classes `Test*` (without `__init__`).
2. Common commands:

| Command | Does |
|---|---|
| `pytest` | run everything under the current folder |
| `pytest -q` / `-v` | quieter / one line per test |
| `pytest test_x.py::test_name` | one test |
| `pytest -k "discount and not bad"` | tests whose names match the expression |
| `pytest -m "not slow"` | select by mark |
| `pytest -x` / `--maxfail=2` | stop after the first/second failure |
| `pytest --lf` | rerun only last-failed tests |
| `pytest -s` | show `print` output (normally captured) |
| `pytest --pdb` | open the debugger at a failure |
| `pytest --durations=5` | list the slowest tests |

3. Settings: `pyproject.toml` `[tool.pytest.ini_options]` (`testpaths = ["tests"]`, `markers = ["slow: ..."]`, `addopts = "-q"`).

#### 15.4.1 Plain `assert` and assertion rewriting

pytest rewrites `assert` in test files so a failure prints the values and a diff (§15.8 shows one). Helpers:

- `pytest.raises(Exc, match="regex")` — the block must raise `Exc` with a message matching the regex (§14.3).
- `pytest.approx(x)` — float comparison with tolerance (relative 1e-6 default): `0.1 + 0.2 == pytest.approx(0.3)`.
- `pytest.warns(Warning)` — the block must emit a warning.

#### 15.4.2 Fixtures

A **fixture** prepares something a test needs (an object, temp file, DB connection) and optionally cleans up. A test requests it by naming it as a parameter; pytest calls it and passes the result (**dependency injection**).

```python
# conftest.py
@pytest.fixture
def inv():
    """A fresh inventory with two products for every test that asks for it."""
    inventory = Inventory()
    inventory.add("mug", 5)
    inventory.add("pen", 10)
    return inventory

# test_inventory_pytest.py
def test_total_units(inv):
    assert inv.total_units() == 15

def test_remove_reduces_stock(inv):
    inv.remove("pen", 4)
    assert inv.stock == {"mug": 5, "pen": 6}

def test_remove_too_many_raises(inv):
    with pytest.raises(OutOfStock, match="want 6, have 5"):
        inv.remove("mug", 6)
```

1. **Fresh per test by default** (`scope="function"`): `test_remove_reduces_stock` changes its `inv`; the next test still gets the original.
2. **Scopes** (how long one value is shared): `function` (default), `class`, `module`, `package`, `session` (one whole run). Wider scopes for expensive read-only resources (one DB engine per session).
3. **Teardown with `yield`** — code after `yield` runs after the test, even if it fails:

   ```python
   @pytest.fixture
   def audit_log(tmp_path):
       f = (tmp_path / "audit.txt").open("w")
       yield f          # the test runs here
       f.close()        # cleanup
   ```

4. Fixtures can use other fixtures (`audit_log` uses `tmp_path`).
5. `autouse=True` applies a fixture to every test in scope without naming it — sparingly.
6. **Built-in fixtures**: `tmp_path` (unique temp directory as a `Path`), `monkeypatch` (temporarily set attributes, env vars, dict items; auto-undone), `capsys` (capture stdout/stderr), `caplog` (capture log records), `request` (info about the requesting test).

### 15.4.3 `tmp_path` — real files, safely

```python
def test_save_and_load_roundtrip(inv, tmp_path):
    path = tmp_path / "stock.json"          # unique temp dir per test
    inv.save(path)
    assert Inventory.load(path) == inv
    assert path.read_text() == '{"mug": 5, "pen": 10}'
```

Each test gets its own directory — no collisions, nothing written into your project. Often better than mocking the file system: it tests the real code path and is still fast.

### 15.4.4 `parametrize` — one test, many cases

```python
@pytest.mark.parametrize(
    ("price", "percent", "expected"),
    [(100, 0, 100), (100, 25, 75), (19.99, 10, 17.99), (50, 100, 0)],
)
def test_discount(price, percent, expected):
    assert apply_discount(price, percent) == pytest.approx(expected)

@pytest.mark.parametrize(("raw", "slug"), [
    ("Blue Mug", "blue-mug"),
    ("  --Hello__World-- ", "hello-world"),
    ("", ""),
], ids=["simple", "punctuation", "empty"])
def test_slugify(raw, slug):
    assert slugify(raw) == slug
```

Each tuple is a **separate test** with its own ID, so one failure doesn't hide others. Include edge cases: empty input, zero, boundaries (0 and 100), negatives, Unicode (`é`, `中`, emoji).

### 15.4.5 Marks

A **mark** labels a test to change how it's selected or run.

| Mark | Effect |
|---|---|
| `@pytest.mark.skip(reason=...)` | never run |
| `@pytest.mark.skipif(sys.platform == "win32", reason=...)` | run only when the condition is false |
| `@pytest.mark.xfail(reason=..., strict=True)` | expected to fail; with `strict=True`, an unexpected *pass* is reported as a failure, so you notice when the bug is fixed |
| `@pytest.mark.parametrize(...)` | data-driven cases |
| `@pytest.mark.slow` (custom) | a label you select with `-m`; register it to avoid "unknown mark" warnings |

```python
@pytest.mark.slow
def test_slow_path():
    time.sleep(0.2)
    assert True

@pytest.mark.xfail(reason="float rounding: 2.675 is stored as 2.67499...; use Decimal",
                   strict=True)
def test_known_bug():
    assert apply_discount(2.675, 0) == 2.68
```

The xfail documents a real float issue: `2.675` is stored as slightly less, so `round(2.675, 2)` is `2.67`; the fix would be `decimal.Decimal`.

### 15.4.6 `conftest.py`

pytest auto-loads `conftest.py`. Its fixtures and hooks (specially named functions pytest calls, e.g. `pytest_configure`) are available to every test in that directory and below — **no import needed**; nested folders can have their own. Ours defines `inv` and registers `slow`:

```python
def pytest_configure(config):
    config.addinivalue_line("markers", "slow: marks a test as slow")
```

(Real projects register markers in `pyproject.toml`.)

#### Output

`pytest -v test_inventory_pytest.py`:

```
test_inventory_pytest.py::test_total_units PASSED                        [  6%]
test_inventory_pytest.py::test_remove_reduces_stock PASSED               [ 12%]
test_inventory_pytest.py::test_remove_too_many_raises PASSED             [ 18%]
test_inventory_pytest.py::test_discount[100-0-100] PASSED                [ 25%]
...
test_inventory_pytest.py::test_discount_rejects_bad_percent[101] PASSED  [ 56%]
test_inventory_pytest.py::test_slugify[simple] PASSED                    [ 62%]
...
test_inventory_pytest.py::test_slow_path PASSED                          [ 93%]
test_inventory_pytest.py::test_known_bug XFAIL (float rounding: 2.67...) [100%]

======================== 15 passed, 1 xfailed in 0.31s ========================
```

`pytest -q -m "not slow" test_inventory_pytest.py` → `14 passed, 1 deselected, 1 xfailed`. `pytest --durations=3` (whole folder):

```
0.20s call     test_inventory_pytest.py::test_slow_path
0.12s call     test_properties.py::test_stock_never_negative
0.04s call     test_properties.py::test_slugify_is_idempotent
```

### Common mistakes

- Tests depending on each other's side effects or order (use fresh fixtures).
- `==` on floats (use `pytest.approx`).
- `pytest.raises` wrapping too much — the exception may come from a setup line.
- Module-level mutable state shared between tests.
- Same-named test files in different folders without `__init__.py` → import-mismatch error.

### In the interview

"I use pytest: plain asserts, fixtures for setup with `yield` for teardown, `parametrize` for edge cases, `tmp_path` and `monkeypatch` for files and environment, and marks to separate slow tests. Shared fixtures go in `conftest.py`."

- *Fixture scope?* Per function by default for isolation; session for expensive read-only resources.
- *xfail vs skip?* Skip = don't run; xfail = run, expect failure (strict flags an unexpected pass).

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## 15.5 Test doubles and `unittest.mock`

### What it is

A **test double** stands in for a real dependency (a network API, a payment gateway, the clock) during a test — like a stunt double. Standard names (Gerard Meszaros' *xUnit Test Patterns*, popularised by Martin Fowler):

| Double | What it does | Example |
|---|---|---|
| **Dummy** | passed but never used | `None` for an unused logger argument |
| **Stub** | returns canned (prepared) answers | `get_rate` always returns 83.5 |
| **Spy** | records how it was called, checked afterwards | a list that collects sent emails |
| **Mock** | pre-programmed with expectations about calls, which the test verifies | "`get_rate` must be called once with `'INR'`" |
| **Fake** | a working, simplified implementation | an in-memory dict instead of a database |

`unittest.mock.Mock` can play stub, spy and mock at once.

### Why it matters

Real dependencies make tests **slow** (network), **flaky** (pass/fail with no code change, e.g. service down), **costly** (paid APIs) or **dangerous** (real emails, real charges), and some cases (a timeout on demand) are hard to trigger.

## How it works — `Mock`

1. `m = Mock()` — any attribute returns another `Mock`; calls are recorded.
2. `m.get_rate.return_value = 83.5` — **stub** the result.
3. `m.get_rate.side_effect = ...` — an **exception** (raised), an **iterable** (one value per call) or a **function** (called with the same arguments).
4. Verify: `assert_called_once_with(...)`, `assert_called_with`, `assert_not_called`, `call_count`, `call_args`, `call_args_list`.
5. `MagicMock` also supports dunder methods (`__len__`, `__iter__`, `__enter__`…); `AsyncMock` (3.8+) for `async def`.
6. **Autospec** — `create_autospec(RealClass, instance=True)` or `patch(..., autospec=True)` gives a mock with only the real object's attributes and signatures. A plain `Mock` accepts misspelled methods, so a test can pass while real code would crash.

```python
class FakeRates:                              # hand-written fake
    def __init__(self, rates): self.rates = rates
    def get_rate(self, currency): return self.rates[currency]

def test_with_fake():
    assert price_in("EUR", 10, FakeRates({"EUR": 0.9})) == 9.0

def test_with_mock_stub_and_verify():
    client = Mock()
    client.get_rate.return_value = 83.5          # stub the answer
    assert price_in("INR", 2, client) == 167.0
    client.get_rate.assert_called_once_with("INR")   # verify the interaction

def test_side_effect_exception():
    client = Mock()
    client.get_rate.side_effect = TimeoutError("rate service down")
    with pytest.raises(TimeoutError):
        price_in("EUR", 10, client)

def test_side_effect_sequence():
    client = Mock()
    client.get_rate.side_effect = [0.9, -1.0]     # one value per call
    assert price_in("EUR", 10, client) == 9.0
    with pytest.raises(ValueError, match="bad rate"):
        price_in("EUR", 10, client)
    assert client.get_rate.call_args_list == [call("EUR"), call("EUR")]

def test_autospec_catches_typos():
    client = create_autospec(FakeRates, instance=True)
    with pytest.raises(AttributeError):
        client.get_rates("EUR")                   # misspelled: plain Mock would allow it
```

## How it works — `patch`

When code creates or imports its dependency itself, `patch` swaps a name **temporarily**:

```python
def test_patch_where_it_is_used(tmp_path):
    # inventory.py did `import json`; patch the name inventory.json.dumps
    with patch("inventory.json.dumps", return_value="{}") as fake_dumps:
        inventory.Inventory({"a": 1}).save(tmp_path / "x.json")
    fake_dumps.assert_called_once()
    assert (tmp_path / "x.json").read_text() == "{}"
```

1. `patch("module.name")` replaces the attribute for a `with` block, a decorated test, or between `start()` / `stop()`; the original is always restored.

2. **Patch where the name is looked up, not where it is defined.** If `shop.py` does `from requests import get`, patch `"shop.get"` — `"requests.get"` won't affect the name already bound in `shop`. (Here `inventory.json` *is* the `json` module, so the patch affects `json.dumps` everywhere during the block — keep patches short.)
3. `patch.object(obj, "attr")` for an object you hold; `patch.dict(os.environ, {...})` for dict entries.
4. pytest's `monkeypatch` covers simple cases: `monkeypatch.setenv("APP_ENV", "test")`, `monkeypatch.setattr(module, "name", value)`.

`pytest -v test_mocking.py` → all 7 tests (the six above plus `test_monkeypatch_env`) `PASSED`.

## Trade-offs

| Approach | Pros | Cons |
|---|---|---|
| Real dependency (tmp_path, SQLite in memory, local server) | tests the real thing | slower, needs setup |
| Fake | realistic behaviour, reusable, readable | you must write and maintain it |
| Mock/stub | quick, can simulate errors/timeouts | couples tests to call details; can drift from the real API |
| `patch` | works without changing the code | fragile if imports move; hints that dependency injection would be cleaner |

### When not to mock

- What you own and can run cheaply (pure functions, dataclasses, `tmp_path` files, in-memory SQLite).
- The thing you're testing.
- Every internal call — that locks in the implementation. Mock at the **boundary** (network, clock, randomness, third-party SDKs) and prefer checking outcomes over interactions.

## Common mistakes

- Patching where defined instead of where used.
- Plain `Mock()` hiding typos — use `autospec`/`spec`.
- Misspelled assertions (`m.called_once_with(...)`, `m.assret_called()`) used to silently "pass" by creating child mocks. Mock now raises `AttributeError` for assertion-like typos (`assret_` etc. since 3.5; names missing the `assert_` prefix since 3.12) unless `unsafe=True`.
- Tests that only prove the mock returns what you told it to.

## Real-world examples

`responses` / `respx` fake HTTP for requests / httpx; `freezegun` / `time-machine` freeze the clock; `moto` fakes AWS (Amazon Web Services) services; `fakeredis` fakes Redis (an in-memory key-value store often used as a cache).

## In the interview

"I prefer dependency injection so I can pass a fake. At external boundaries I use `unittest.mock` with autospec: `return_value` to stub, `side_effect` for errors or sequences, `assert_called_once_with` only when the interaction matters. With `patch` I patch where the name is looked up."

- *Mock vs stub?* A stub supplies answers; a mock also verifies how it was called.
- *Why autospec?* The mock fails on methods or signatures the real object lacks.
- *`side_effect = [1, 2]`, three calls?* 1, then 2, then `StopIteration`.

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## 15.6 Coverage

### What it is

**Code coverage** measures which lines (and branches) ran during the tests. `coverage.py` is the standard tool; `pytest-cov` integrates it.

- **Line (statement) coverage** — did each line run?
- **Branch coverage** — did each `if` go both ways? A line can run while one side is never taken.

### How it works

```
pytest --cov=inventory --cov-branch --cov-report=term-missing
# or without the plugin:
coverage run --branch -m pytest && coverage report -m && coverage html   # htmlcov/index.html
```

Output:

```
Name           Stmts   Miss Branch BrPart  Cover   Missing
-----------------------------------------------------------
inventory.py      48      7     10      2    84%   48, 81-86
-----------------------------------------------------------
TOTAL             48      7     10      2    84%
```

48 statements, 7 never ran; 10 branch points, 2 partially taken. Line **48** is `raise ValueError("qty must be positive")` in `Inventory.add` — no test passes a non-positive quantity, so the validation is untested. **81-86** is the `if __name__ == "__main__":` demo block (exclude with a `# pragma: no cover` comment or config).

### Trade-offs / how to use the number

- Coverage shows what is **not** tested, never that tested code is **correct** — a line run without any assertion still "counts".
- Teams often enforce a CI threshold (`--cov-fail-under=85`). 100% is rarely worth it outside small critical libraries; aim high on business logic and error paths.
- In code review, read the *missing* list more than the percentage.
- **Mutation testing** (`mutmut`) goes further: it mutates code (`<` → `<=`) and checks some test fails.

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## 15.7 doctest — examples that are tests

`doctest` finds REPL-style examples in docstrings (`>>> expr` then expected output) and checks that running them produces that output.

```
def slugify(name: str) -> str:
    """Lower-case, keep letters/digits, join words with '-'.

    >>> slugify("Hello, World!")
    'hello-world'
    """
```

- Run: `python -m doctest inventory.py -v`, `doctest.testmod()` in the module, or `pytest --doctest-modules`.
- Output is compared as **text**, so it must match the `repr` exactly.

Output of `python inventory.py` and `pytest --doctest-modules inventory.py`:

```
stock: {'mug': 3} | units: 3
doctest: TestResults(failed=0, attempted=3)
..                                                                       [100%]
2 passed in 0.07s
```

(3 examples in 2 docstrings; pytest counts each docstring as one test.)

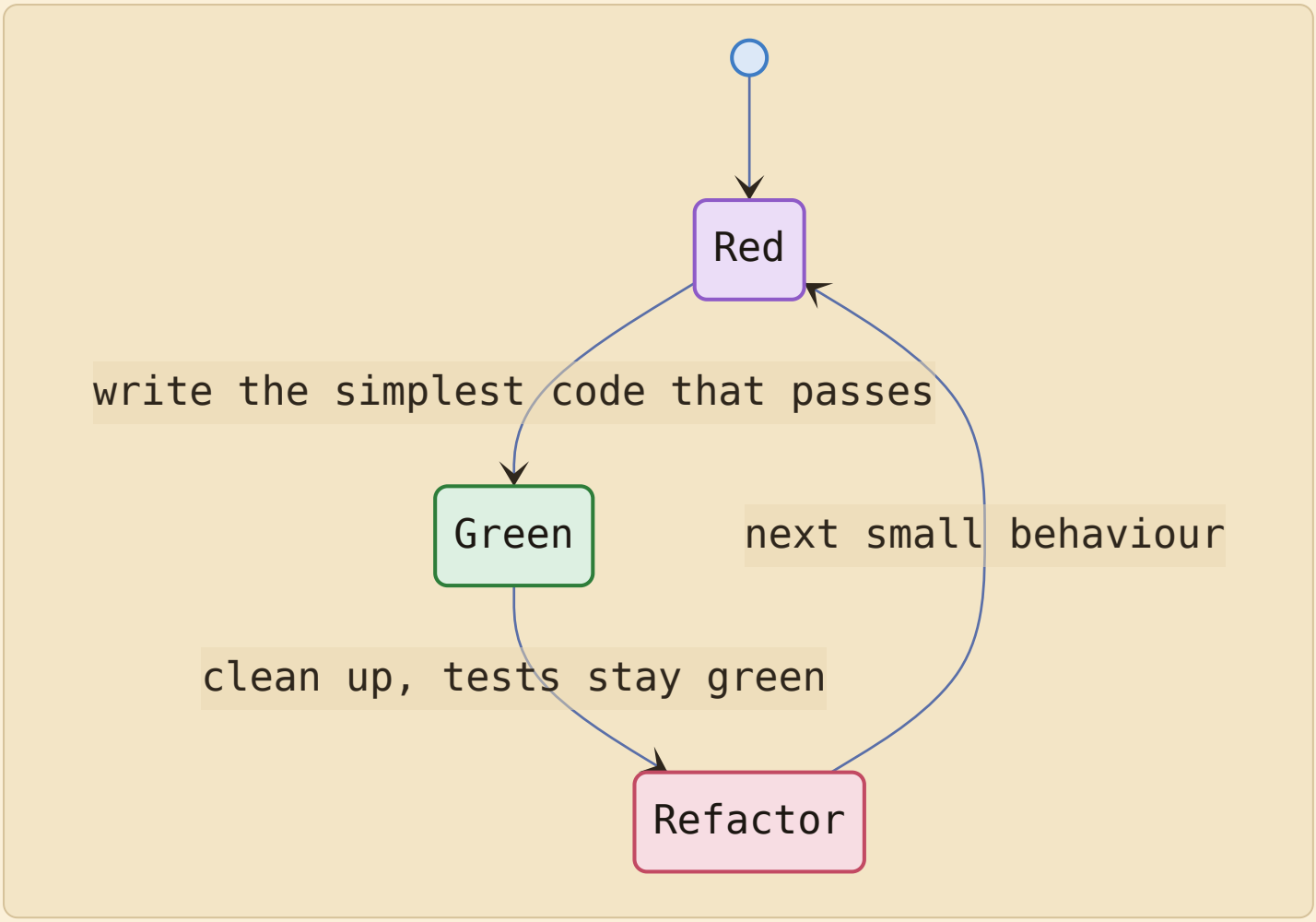
**Trade-off:** great for keeping **documentation examples correct**; poor as a test suite — text comparison is brittle (dict order, float digits, memory addresses) and there are no fixtures or parametrization. Doctests for docs, pytest for tests.

---

## 15.8 The TDD loop (red, green, refactor)

### What it is

**Test-driven development** (popularised by Kent Beck): write the test **before** the code, in tiny cycles.




1. **Red** — a small failing test for the next behaviour; see it fail *for the right reason*.
2. **Green** — the minimum code to pass.
3. **Refactor** — improve names and structure under the tests' protection.

### How it works — a real cycle

Feature: `Inventory.low_stock(threshold)` → SKUs (stock-keeping units, product codes) with quantity **at or below** the threshold, sorted.

**Red.** Tests first (`test_tdd_low_stock.py`):

```python
def test_low_stock_lists_skus_below_threshold(inv):      # inv: mug=5, pen=10
    assert inv.low_stock(threshold=6) == ["mug"]

def test_low_stock_sorted_and_inclusive(inv):
    inv.add("cap", 1)
    assert inv.low_stock(threshold=10) == ["cap", "mug", "pen"]
```

Both fail — the method doesn't exist:

```
>       assert inv.low_stock(threshold=6) == ["mug"]
               ^^^^^^^^^^^^^
E       AttributeError: 'Inventory' object has no attribute 'low_stock'
...
2 failed in 0.07s
```

**Green, first try.** `return [sku for sku, q in self.stock.items() if q < threshold]` passes the first test only; assertion rewriting shows why:

```
>       assert inv.low_stock(threshold=10) == ["cap", "mug", "pen"]
E       AssertionError: assert ['mug', 'cap'] == ['cap', 'mug', 'pen']
E
E         At index 0 diff: 'mug' != 'cap'
E         Right contains one more item: 'pen'
E         Use -v to get more diff
1 failed, 1 passed in 0.07s
```

Two bugs: not sorted, and `pen` (exactly 10) is missing, so the comparison must be `<=`.

**Green.**

```python
def low_stock(self, threshold: int) -> list[str]:
    """SKUs with quantity <= threshold, alphabetically."""
    return sorted(sku for sku, qty in self.stock.items() if qty <= threshold)
```

Whole suite: `31 passed, 1 skipped, 1 xfailed`.

**Refactor.** Names are already clear — next behaviour.

### Trade-offs

| Pros | Cons |
|---|---|
| Every line has a reason (a test) | Slower start; feels unnatural at first |
| Forces small, testable design | Hard when exploring an unknown problem/API (spike — a throw-away experiment — first, then test) |
| Instant regression safety | Can over-fit tests to one design if cycles are too big |

Many teams use TDD at least for bug fixes: **first write a test that reproduces the bug**, see it fail, then fix — so it can never silently return.

---

## 15.9 Property-based testing with Hypothesis (brief)

### What it is

In an **example-based** test you pick inputs. In a **property-based** test you state a rule that must hold for *all* inputs of a kind, and the library generates hundreds — including nasty ones (empty strings, huge numbers, Unicode, NaN) — to break it. **Hypothesis** is Python's main library (the idea comes from Haskell's QuickCheck). A failing input is **shrunk** to the simplest failing example and saved in `.hypothesis/` so it's retried first next time.

### How it works

```python
from hypothesis import given, settings, strategies as st

@given(st.text())
def test_slugify_is_idempotent(s):            # applying it twice changes nothing
    assert slugify(slugify(s)) == slugify(s)

@given(st.text())
def test_slug_has_only_safe_chars(s):
    assert all(c.isascii() and (c.isalnum() or c == "-") for c in slugify(s))

@given(price=st.floats(0, 1e6), pct=st.floats(0, 100))
def test_discount_never_increases_price(price, pct):
    assert apply_discount(price, pct) <= round(price, 2)

@settings(max_examples=200)
@given(st.lists(st.tuples(st.sampled_from(["add", "remove"]), st.integers(1, 20))))
def test_stock_never_negative(ops):           # random sequences of operations
    inv = Inventory()
    for op, qty in ops:
        try:
            getattr(inv, op)("mug", qty)
        except OutOfStock:
            pass
        assert inv.stock.get("mug", 0) >= 0
```

All four pass. Against a naive `name.strip().lower().replace(" ", "-")`, the "safe characters" property fails and shrinks to one character:

```
E       Failing test case: test_slug_has_only_safe_chars(
E           s=':',
E       )
```

Common properties: **round-trip** (`load(save(x)) == x`, `decode(encode(x)) == x`), **idempotence** (`f(f(x)) == f(x)`), **invariants** (stock never negative, output sorted, length preserved), **oracle** (fast version equals a slow obviously-correct one). Great for parsers and serializers.

**When not to use it:** no general rule to state, or inputs expensive to build. Use alongside example tests, not instead.

---

## 15.10 Reading tracebacks

### What it is

A **traceback** reports an uncaught exception: the **call stack** that led to it, oldest call at the top, failing line at the bottom.

### How it works — read it bottom-up

1. **Last line first** — type and message: `ValueError: invalid literal for int() with base 10: 'two'`.
2. **The frame above it** (each entry is a *frame* — one function call in progress): file, line, function. Since 3.11, `~~~^^^` markers point at the exact sub-expression (PEP 657).
3. **Walk upward** to *your* code (skip library frames) — often the bug is where your code passed bad data in.
4. **Chained exceptions**: "The above exception was the direct cause of the following exception" = explicit `raise NewError(...) from original`. "During handling of the above exception, another exception occurred" = a new error *inside* an `except` block (implicit chaining).

From `debugging_demo.py` (`"two"` as a quantity):

```
Traceback (most recent call last):
  File "debugging_demo.py", line 19, in load_orders
    orders.append(parse_order(line))
                  ~~~~~~~~~~~^^^^^^
  File "debugging_demo.py", line 12, in parse_order
    return {"sku": sku, "qty": int(qty), "price": float(price)}
                               ~~~^^^^^
ValueError: invalid literal for int() with base 10: 'two'

The above exception was the direct cause of the following exception:

Traceback (most recent call last):
  File "debugging_demo.py", line 35, in <module>
    load_orders(["mug,2,4.50", "pen,two,1.20"])
    ~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "debugging_demo.py", line 21, in load_orders
    raise RuntimeError(f"bad order on line {n}: {line!r}") from e
RuntimeError: bad order on line 2: 'pen,two,1.20'
```

The final error says *which input* was bad; the cause says *what* was wrong (`int('two')`) — the payoff of `raise ... from e` with context.

**Common exception → usual cause**

| Exception | Usual cause |
|---|---|
| `NameError` | typo, or used before assignment |
| `AttributeError: 'NoneType' object has no attribute ...` | a function returned `None` (forgot `return`, or an in-place method like `list.sort()`) |
| `TypeError: ... takes 2 positional arguments but 3 were given` | forgot `self`, or wrong call |
| `KeyError` / `IndexError` | missing dict key / off-by-one |
| `UnboundLocalError` | assigning to a name inside a function makes it local (use `nonlocal`/`global` or rename) |
| `ModuleNotFoundError` | wrong virtual environment (venv), or a local file shadows the module (your own `random.py`) |
| `RecursionError` | missing or unreachable base case |

---

## 15.11 Debugging with `pdb` and `breakpoint()`

### What it is

`pdb` is Python's interactive **debugger**: pause at a line, inspect variables, step, move up and down the stack. `breakpoint()` (3.7+, PEP 553) pauses there — it calls `pdb.set_trace()` by default.

## How it works — ways in

| Way | When |
|---|---|
| Put `breakpoint()` in the code | pause at a specific spot |
| `python -m pdb script.py` | start the whole script under the debugger |
| `pdb.pm()` after a crash (in the REPL) or `python -m pdb` (enters post-mortem — inspecting the state at the moment of the crash — on an uncaught exception) | inspect the frame where it died |
| `pytest --pdb` | debugger at the failing assertion |
| `pytest --trace` | debugger at the start of each test |
| IDE (integrated development environment: VS Code, PyCharm) breakpoints | same ideas, with a GUI |

`PYTHONBREAKPOINT` controls `breakpoint()`: `=0` disables all (safety net in CI), `=ipdb.set_trace` switches debugger.

### The commands you need

| Command | Does |
|---|---|
| `l` / `ll` | list code around the current line / the whole function |
| `n` (next) | run the current line, stepping *over* calls |
| `s` (step) | step *into* a call |
| `r` (return) | run until the current function returns |
| `c` (continue) | run until the next breakpoint |
| `unt N` (until) | run until line N |
| `p expr` / `pp expr` | print / pretty-print an expression |
| `w` (where) | show the call stack |
| `u` / `d` | move up / down the stack to inspect callers |
| `b file:line` / `b func` / `b line, cond` | set a (conditional) breakpoint |
| `cl N` | clear breakpoint N |
| `display expr` | show `expr` each time execution stops if it changed |
| `interact` | a full Python REPL in the current scope |
| `q` | quit |

A bare expression works too, but for variables named like commands (`n`, `c`, `l`…) use `p n`.

### A real session (scripted, so it can be shown)

`debugging_demo.py` feeds `pdb` a command list. The function crashes on an empty list:

```python
def average_value(orders: list[dict]) -> float:
    total = 0.0
    for o in orders:
        total += o["qty"] * o["price"]
    return total / len(orders)          # bug when orders is empty
```

Commands: `b average_value`, `c`, `p orders`, `n`, `n`, `p total, len(orders)`, `c`. Transcript:

```
(Pdb) Breakpoint 1 at debugging_demo.py:26
(Pdb) > debugging_demo.py(26)average_value()
-> total = 0.0
(Pdb) []
(Pdb) > debugging_demo.py(27)average_value()
-> for o in orders:
(Pdb) > debugging_demo.py(29)average_value()
-> return total / len(orders)          # bug when orders is empty
(Pdb) (0.0, 0)
(Pdb)
then it raised: ZeroDivisionError('division by zero')
post-mortem locals: {'orders': [], 'total': 0.0}
breakpoint() skipped because PYTHONBREAKPOINT=0
```

The loop body is skipped and `len(orders)` is 0 at the return — handle the empty case (`if not orders: return 0.0` or raise a clear error). "post-mortem locals" is the same information recovered from the innermost traceback frame after the crash — what `pdb.pm()` gives interactively.

### A debugging method (not just a tool)

1. **Reproduce** reliably — ideally as a failing test.
2. **Read** the traceback and code; state a hypothesis.
3. **Observe** with `breakpoint()`, logging or `p`.
4. **Narrow down**: halve the input, comment out half the code, or `git bisect` (binary-searches history for the commit that introduced the bug).
5. **Fix** the cause, not the symptom; keep the test.
6. Explaining the code line by line ("rubber-duck debugging") often reveals the bug.

#### Common mistakes

- Committing a `breakpoint()` — CI hangs waiting for input. Set `PYTHONBREAKPOINT=0` there and use a linter rule such as Ruff's `T100`.
- Changing several things at once.

---

## 15.12 Logging vs `print` for debugging

`print` is fine for a five-minute look. Anything that stays in the code uses **logging** (Chapter 14) — levels, timestamps, module names, configurable destinations, DEBUG enabled only where needed.

| | `print` | `logging` |
|---|---|---|
| Levels / filtering | no | DEBUG…CRITICAL, per module |
| Timestamp, module, line | manual | via formatter |
| Destination | stdout | console, files, syslog (the OS's central log), log services |
| Safe to leave in code | no | yes (DEBUG off in production) |
| In tests | captured by pytest (`-s` shows) | `caplog` fixture lets tests assert on logs |

```
log = logging.getLogger(__name__)
log.debug("pricing order=%s rate=%s", order_id, rate)    # visible only at DEBUG
```

Tips: raise one module's level temporarily (`logging.getLogger("shop.pricing").setLevel(logging.DEBUG)`); `print(f"{total=}")` prints `total=0.0`; `python -X dev` enables extra runtime checks and warnings; `faulthandler` (`python -X faulthandler`) dumps tracebacks on hard crashes (e.g. a segfault in C code) or hangs.

## 15.13 Common bugs checklist

Run through this when something "can't possibly be wrong":

| # | Bug | Example | Fix |
|---|---|---|---|
| 1 | Mutable default argument | `def f(items=[])` shares one list across calls | `items=None` then `items = [] if items is None else items` |
| 2 | Late-binding closures | `[lambda: i for i in range(3)]` all return 2 | `lambda i=i: i` |
| 3 | Aliasing / shallow copies | `grid = [[0]*3]*3` — one row shared three times | `[[0]*3 for _ in range(3)]`, `copy.deepcopy` |
| 4 | Modifying a list while iterating | removing items inside `for x in xs` skips items | iterate over a copy or build a new list |
| 5 | `is` vs `==` | `x is 1000` depends on object identity | `==` for values; `is` only for `None`, sentinels (unique marker objects) |
| 6 | Float equality | `0.1 + 0.2 == 0.3` is False | `math.isclose`, `pytest.approx`, `Decimal` for money |
| 7 | Integer vs float division | `7 / 2 == 3.5`, `7 // 2 == 3`, `-7 // 2 == -4` | pick the right operator |
| 8 | Methods returning `None` | `xs = xs.sort()` makes `xs` None | `xs.sort()` or `xs = sorted(xs)` |
| 9 | Swallowed exceptions | `except Exception: pass` hides the real error | catch specific exceptions, log, re-raise |
| 10 | Off-by-one | `range(1, n)` excludes `n` | test the boundaries |
| 11 | Truthiness | `if count:` is False for `0` | `if count is not None:` |
| 12 | Naive datetimes / time zones | comparing local and UTC times | aware UTC datetimes (Chapter 14) |
| 13 | Unclosed resources | files/connections left open | `with` statements |
| 14 | Wrong environment | `ModuleNotFoundError` though "it's installed" | check `sys.executable`, activate the venv, `python -m pip` |
| 15 | Module name shadowing | your `random.py` or `test.py` hides the stdlib | rename the file |
| 16 | Global/shared state between tests | tests pass alone, fail together | fresh fixtures, no module-level mutable state |
| 17 | Encoding | `UnicodeDecodeError` reading files | `open(p, encoding="utf-8")` |
| 18 | Unseeded randomness / real clock in tests | flaky tests | seed the generator, inject the clock, freeze time |

---

## 15.14 Chapter self-check

1. Three parts of a well-shaped test? → Arrange, Act, Assert.
2. Why do most tests belong at the bottom of the pyramid? → Unit tests are fast, precise, cheap.
3. How does pytest show values in a failing `assert`? → Assertion rewriting.
4. How does a test get a fixture, and must it import from `conftest.py`? → By naming it as a parameter; no, discovered automatically.

5. Fixture cleanup? → Code after `yield`.
6. `parametrize` with four tuples produces? → Four separate tests.
7. What does `tmp_path` give you? → A unique temporary directory (`Path`) per test.
8. `xfail(strict=True)` and the test passes? → Reported as a failure (XPASS strict), so you notice the fix.
9. Stub vs mock vs fake? → Canned answers / verified interactions / working simplified implementation (e.g. an in-memory repository).
10. `shop.py` has `from requests import get` — what do you patch? → `"shop.get"`.
11. Why `create_autospec`? → Mocks reject attributes and signatures the real object lacks.
12. What does 100% line coverage prove? → Only that every line ran — not that results were checked.
13. TDD cycle? → Red, green, refactor.
14. What does Hypothesis do after finding a failing input? → Shrinks it to a minimal example and remembers it.
15. How do you read a traceback? → Bottom-up: message, failing line, then up to your code; "direct cause" = `from`, "during handling" = error inside `except`.
16. Disable every `breakpoint()` in CI? → `PYTHONBREAKPOINT=0`.
17. `xs = xs.sort()` — what is `xs`? → `None`.

## Chapter TL;DR

- **Why:** tests make change safe. Many fast unit tests, fewer integration/E2E; deterministic, isolated, behaviour-focused; run in CI.
- **unittest:** `TestCase`, `setUp`, `assertEqual`/`assertRaises`; built in, more verbose.
- **pytest:** plain `assert` (rewritten failure diffs), fixtures (`yield` teardown, scopes, `conftest.py`), `parametrize`, `tmp_path`, `monkeypatch`, `capsys`, `caplog`, marks (`skip`, `skipif`, `xfail`, custom + `-m`), `-k`, `-x`, `--lf`, `--pdb`.
- **Doubles:** dummy, stub, spy, mock, fake. `Mock` + `return_value`/`side_effect`/`assert_called_once_with`; `patch` where the name is used; autospec; mock only at boundaries; prefer dependency injection.
- **Coverage:** `pytest --cov --cov-branch --cov-report=term-missing` finds untested code; it doesn't prove correctness.
- **TDD:** red, green, refactor; always reproduce a bug with a failing test first.
- **doctest** keeps doc examples true; **Hypothesis** checks properties and shrinks counter-examples.
- **Debugging:** read tracebacks bottom-up; `breakpoint()`/`pdb` (`n s c p w u d b`), `pytest --pdb`, post-mortem; reproduce → hypothesise → observe → narrow → fix → keep the test.
- **Logging over print** for anything that stays; check the **common bugs list** first (mutable defaults, aliasing, `None` returns, float equality, swallowed exceptions, wrong venv).